

Econometrics

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Lecture 1 Notes

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1 Introduction

1.1 Definition of Econometrics

This is the social science in which the tools of economic theory, mathematics and statistical inference are applied to the analysis of economic phenomena.

1.2 Types of Econometrics

1.2.1 Theoretical Econometrics

This is concerned with the development of appropriate methods for measuring economic relationships specified by econometric models. Relies heavily on mathematical statistics

1.2.2 Applied Econometrics

Here we use the tools of theoretical econometrics to study some special fields of economics and business such demand and supply.

1.3 Methodology of Econometrics

- Statement of theory or hypothesis
- Specification of mathematical model of the theory
- Specification of statistical or econometric model
- Obtaining the data
- Estimating parameters of econometric model
- Hypothesis testing
- Forecasting or prediction
- Using model for control or policy purposes

1.4 Structure of Economic Data

1.4.1 Cross sectional Data

These consists of a sample of individuals, households, firms, states, countries taken at a given point in time.

Example may include a data set on wages and other individual features.

Obs No	wage	educ	exper	gender	married
1	3.10	11	2	1	0
2	3.24	12	32	0	0
3	5.30	4	44	1	

1.4.2 Time series data

A time series data set consists of observations on a variable or several variables made sequentially over time.

Examples include:

- Stock prices
- GDP
- Automobile sales figures
- CPI

Obs	Year	GDP
1	2006	14613.8

Obs	Year	GDP
2	2007	14873.8
3	2008	14830.4

1.4.3 Panel data

A panel data consist of a time series for each cross sectional member in a data set.

The key aspect of panel data is that we observe each micro unit for a number of time periods

Firm	Year	Area	Prod	Labor
1	1990	7.87	2.50	160
1	1991	7.18	2.50	151
1	1992	7.20	2.50	138
2	1990	8.92	2.50	184
2	1991	7.31	2.50	161
2	1992	8.01	2.50	151

2 Regression Analysis with Cross Sectional Data

2.1 Simple Regression Model

The simple regression model can be used to study the relationship between two variables.

A random experiment is repeated n times under identical conditions.

For each trial $i = 1, 2, \dots, n$ the value of X_i is known and the response Y_i is recorded.

The simple linear regression model is given by:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i \quad (1)$$

2.1.1 Deriving the OLS

Given

$$Y_i = \beta_0 + \beta_1 X_i$$

The sum of squared errors:

$$Q = \sum \epsilon_i^2 = \sum (Y_i - \beta_0 - \beta_1 X_i)^2$$

Differentiating w.r.t β_0 and β_1 we have:

$$\frac{\partial Q}{\partial \beta_0} = -2 \sum (Y_i - \beta_0 - \beta_1 X_i)$$

$$\frac{\partial Q}{\partial \beta_1} = -2 \sum X_i (Y_i - \beta_0 - \beta_1 X_i)$$

We can expand the equations and have:

$$\sum Y_i = n\hat{\beta}_0 + \hat{\beta}_1 \sum X_i \quad (2)$$

$$\sum X_i Y_i = \hat{\beta}_0 \sum X_i + \hat{\beta}_1 \sum X_i^2 \quad (3)$$

Solving the above simultaneously we have:

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

$$\hat{\beta}_1 = \frac{n \sum X_i Y_i - \sum X_i \sum Y_i}{n \sum X_i^2 - (\sum X_i)^2}$$

The fitted values of Y is

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$$

The residuals for observation i is the difference between actual and its fitted value.

$$e_i = Y_i - \hat{Y}_i$$

The total sum of squares denoted by SST

$$SST = \sum (Y_i - \bar{Y})^2$$

The Sum of Squares due to regression:

$$SSR = \sum (\hat{Y}_i - \bar{Y})^2$$

The Sum of Squares due to errors:

$$SSE = \sum (\hat{e}_i)^2$$

This implies that:

$$SST = SSE + SSR$$

To proof:

Expand $\sum (Y_i - \bar{Y})^2$ by adding and subtracting $\hat{Y}_i$

2.1.2 Goodness of Fit

The R-squared of the regression sometimes called the coefficient of determination.

$$R^2 = \frac{SSR}{SST} = 1 - \frac{SSE}{SST}$$

R^2 is the fraction of sample variation in Y that is explained by X .

When interpreting R^2 we multiply by 100. R^2 is the % of sample variation in Y that is explained by X .

2.2 Gauss Markov Theorem

Under the assumptions of Simple Linear Regression, the least square estimators $\hat{\beta}_0$ and $\hat{\beta}_1$ are unbiased and have a minimum variance among all linear unbiased estimators of β_0 and β_1 . Thus $\hat{\beta}_0$ and $\hat{\beta}_1$ are said to be BLUE.

2.3 Linear Estimators

The least squares intercept and the slope are linear estimators in the sense that they are linear function of Y_i

Consider:

$$\hat{\beta}_1 = \frac{\sum(Y_i - \bar{Y})}{\sum(X_i - \bar{X})^2}$$

can be written as:

$$\hat{\beta}_1 = \sum m_i Y_i$$

$$\text{where } m_i = \frac{(X_i - \bar{X})}{\sum(X_i - \bar{X})^2}$$

and $\sum m_i = 0$ and $\sum m_i X_i = 1$

2.3.1 Unbiasedness of estimates

$$\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$$

$$\begin{aligned} E[\hat{\beta}_0] &= E[\bar{Y} - \hat{\beta}_1 \bar{X}] \\ &= \hat{\beta}_0 + \hat{\beta}_1 \bar{X} - \hat{\beta}_1 \bar{X} \\ &= \hat{\beta}_0 \end{aligned} \tag{4}$$

$$\begin{aligned} E[\hat{\beta}_1] &= \sum m_i E[Y_i] \\ &= \sum m_i (\beta_0 + \beta_1 X_i) \\ &= \beta_1 \end{aligned} \tag{5}$$

2.3.2 Variances

$$\begin{aligned} Var(\hat{\beta}_1) &= Var(\sum m_i Y_i) \\ &= \sum m_i^2 Var(Y_i) + \sum \sum k_i k_j cov(Y_i, Y_j) \\ &= \sigma^2 \frac{\sum(X_i - \bar{X})^2}{\sum(X_i - \bar{X})^4} \\ &= \frac{\sigma^2}{\sum(X_i - \bar{X})^2} \end{aligned} \tag{6}$$

$$\begin{aligned} Var(\hat{\beta}_0) &= Var(\bar{Y}) + \bar{X}^2 Var(\hat{\beta}_1) - 2\bar{X} Cov(\bar{Y}, \hat{\beta}_1) \\ &= \frac{\sigma^2}{n} + \bar{X}^2 \frac{\sigma^2}{\sum(X_i - \bar{X})^2} \\ &= \sigma^2 \left(\frac{1}{n} + \frac{\bar{X}^2}{\sum(X_i - \bar{X})^2} \right) \end{aligned} \tag{7}$$

2.3.3 Covariances

The covariance between β_0 and β_1 is

$$\begin{aligned} Cov(\beta_0, \beta_1) &= Cov(\bar{Y}, \beta_1) - \bar{X}Var(\beta_1) \\ &= -\frac{\bar{X}\sigma^2}{\sum(X_i - \bar{X})^2} \end{aligned} \quad (8)$$

2.4 Inference for β_0 and β_1

2.4.1 Inference for β_1

We test the hypothesis concerning β_1 :

$$H_0 : \beta_1 = 0 \text{ vs } H_1 : \beta_1 \neq 0$$

The sampling distribution of $\hat{\beta}_1$ refers to the different values of $\hat{\beta}_1$ that would be obtained with repeated sampling when the levels of the predictor X are held constant from sample to sample

$$E[\hat{\beta}_1] = \beta_1$$

and

$$Var(\hat{\beta}_1) = \frac{\sigma^2}{\sum(X_i - \bar{X})^2}$$

An estimate for σ^2 is:

$$\sigma^2 = \frac{SSE}{n - 2}$$

thus

$$S^2(\hat{\beta}_1) = \frac{MSE}{\sum(X_i - \bar{X})^2}$$

If Y_i are normally distributed then the distribution of $\hat{\beta}_1$ is normal since $\hat{\beta}_1 = \sum m_i Y_i$ and a linear combination of independent random variables are also normally distributed then:

$$\hat{\beta}_1 \sim N(\beta_1, \frac{\sigma^2}{\sum(X_i - \bar{X})^2})$$

The $(1 - \alpha)100\%$ CI for β_1 is

$$\hat{\beta}_1 \pm t_{(1-\frac{\alpha}{2}), n-2} \sqrt{\frac{MSE}{\sum(X_i - \bar{X})^2}}$$

To test the hypothesis $H_0 : \beta_1 = c$ the test statistic is:

$$t = \frac{\hat{\beta}_1 - c}{\sqrt{\frac{MSE}{\sum(X_i - \bar{X})^2}}}$$

2.4.2 Inference for β_0

The sampling distribution of β_0 is:

$$\hat{\beta}_0 \sim N(\beta_0, \sigma^2(\frac{1}{n} + \frac{\bar{x}^2}{\sum(x_i - \bar{x})^2}))$$

The $(1 - \alpha)100\%$ CI for β_0 is

$$\hat{\beta}_0 \pm t_{(1-\frac{\alpha}{2}), n-2} \sqrt{S^2(\hat{\beta}_0)}$$

To test the hypothesis To test the hypothesis $H_0 : \beta_0 = c$ the test statistic is:

$$t = \frac{\hat{\beta}_0 - c}{\sqrt{S^2(\hat{\beta}_0)}}$$

2.5 Practical in R

2.5.1 Example 1: US Consumption Expenditure

In `fpp3` package in R, a data set named `us_change` shows a time series of quarterly percentage changes (growth rates) of real personal consumption expenditure, y and real personal disposable income x for the US from 1970 Q1 to 2019 Q2.

```
us_change %>%
  pivot_longer(c(Consumption, Income), names_to = "Series") %>%
  autoplot(value) + theme_bw() +
  labs(y = "% Change", x = "Time")
```

A scatter plot of consumption changes against income changes:

```
us_change |>
  ggplot(aes(x = Income, y = Consumption)) +
  labs(y = "Consumption (quarterly % change)",
       x = "Income (quarterly % change)") +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  theme_bw()
```

The model can be fitted using:

```
model <- lm(Consumption ~ Income, data = us_change)
pander(summary(model))
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.5445	0.05403	10.08	1.63e-19
Income	0.2718	0.04673	5.817	2.402e-08

Table 5: Fitting linear model: Consumption ~ Income

Observations	Residual Std. Error	R^2	Adjusted R^2
198	0.5905	0.1472	0.1429

The fitted equation is:

$$\hat{Y} = 0.545 + 0.272X$$

ANOVA Table can be obtained using:

```
pander(anova(model))
```

Table 6: Analysis of Variance Table

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Income	1	11.8	11.8	33.84	2.402e-08
Residuals	196	68.35	0.3487	NA	NA

The confidence interval of the model:

```
pander(confint(model))
```

	2.5 %	97.5 %
(Intercept)	0.438	0.6511
Income	0.1797	0.364

2.5.2 Example 2: Sales data

Consider the data frame named `marketing` in the `datarium` package containing the impact of three advertising medias (youtube, facebook and newspaper) on sales. We want to fit a SLR to see the impact of advertising budget spent on youtube on sales.

- Create a visualization for the two variables
- Fit a SLR model
- Obtain the 95% confidence interval and the ANOVA table for the model
- Interpret the results

2.6 Multiple Linear Regression Analysis

The general multiple linear regression model can be written as:

$$Y = \beta_0 + \beta_1 X_1 + \dots + \beta_k X_k + \epsilon$$

Where β_0 is the intercept

$\beta_1, \dots, \beta_k$ are the slope parameters associated with $x_1, \dots, x_k$

2.6.1 Matrix Formulation of MLR

Consider the following multiple regression model:

$$Y_i = \beta_0 + \beta_1 X_{i1} + \beta_2 X_{i2} + \dots + \beta_p X_{i,p-1} + \epsilon_i$$

The model can be written using vectors and matrices as:

$$Y = X\beta + \epsilon$$

$Y = n \times 1$ vector of response values $\beta = p \times 1$ vector of regression parameters $X = n \times p$ matrix of known constants $\epsilon = n \times 1$ vector of *iid* error terms

Interpreting the slope parameters has to take to account possible changes in other independent variables a slope parameter say β_k gives the change in Y when the independent variable X_k changes by one unit while all other independent variables remain constant.

The coefficients measure the *marginal effects* of the predictor variables.

2.6.2 OLS Estimation

Define the best estimate of β as that which minimizes the SSE $\epsilon'\epsilon$

$$\begin{aligned} \sum \epsilon_i^2 &= \epsilon'\epsilon \\ &= (Y - X\beta)'(Y - X\beta) \end{aligned} \tag{9}$$

Differentiate w.r.t β and equate to zero and have:

$$Q = (Y - X\beta)'(Y - X\beta) = Y'Y - 2Y'X\beta + \beta'X'X\beta$$

$$\frac{\partial Q}{\partial \beta} = 2X'X\beta - 2Y'X$$

Equating to zero we have:

$$-2X'Y = -2X'X\beta$$

Solving for β we get:

$$\hat{\beta} = (X'X)^{-1}X'Y$$

2.6.3 Fitted Values and Residuals

The fitted values are given as:

$$\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{i1} + \dots + \hat{\beta}_{p-1} X_{i,p-1}$$

Residuals are given by:

$$e_i = Y_i - \hat{Y}_i$$

In matrix formulation:

$$\begin{aligned}
\hat{Y} &= X\hat{\beta} \\
&= X(X'X)^{-1}X'Y \\
&= HY
\end{aligned} \tag{10}$$

In other words:

$$\begin{aligned}
e &= Y - \hat{Y} \\
&= Y - X\hat{\beta} \\
&= Y - X(X'X)^{-1}X'Y \\
&= Y - HY \\
&= (1 - H)Y
\end{aligned} \tag{11}$$

2.6.4 Inference in MLR

If the model $Y = X\beta + \epsilon$ is correct, the expectation of Y is $X\beta$ and the expectation of β is:

$$\begin{aligned}
E[\hat{\beta}] &= [(X'X)^{-1}X']E[Y] \\
&= [(X'X)^{-1}X']X\beta \\
&= \beta
\end{aligned} \tag{12}$$

The variance of β is

$$\begin{aligned}
Var(\beta) &= [(X'X)^{-1}]Var(Y)[(X'X)^{-1}X']' \\
&= [(X'X)^{-1}X']I\sigma[(X'X)^{-1}X']' \\
&= \sigma^2(X'X)^{-1}
\end{aligned} \tag{13}$$

The ANOVA table can be constructed as:

Source	df	SS	MSS
Regression	p-1	SSR	MSR
Error	n-p	SSE	MSE
Total	n-1	SST	

2.6.5 Coefficient of Multiple Determination

$$R^2 = \frac{SSR}{SST}$$

It measures the amount of variation in Y explained by the independent variables.

The adjusted R^2 is given by:

$$R^2_{\alpha} = 1 - \frac{(n-1)SSE}{(n-p)SST}$$

It adjusts the R^2 for the number of predictors in the model.

2.6.6 Hypothesis testing for individual regressors

- Determine the null and alternative hypothesis
- Specify the test statistic and its distribution if H_0 is true
- Select α and determine the rejection region
- Calculate the sample value of test statistic and desired p-value
- State your conclusion

The hypothesis is

$$H_0 : \beta_k = 0 \text{ vs } H_1 : \beta_k \neq 0$$

The test statistic is:

$$t = \frac{\beta_k}{se(\beta_k)} \sim t_{n-p}$$

2.6.7 Global test

This is an overall test for the regression model. It investigates the possibility that all the regression coefficients are equal to zero.

$$H_0 : \beta_1 = \dots = \beta_k = 0 \text{ vs } H_a : \beta_j \neq 0$$

The test statistic is the F -statistic given by

$$F = \frac{MSR}{MSE}$$

2.6.8 Assumptions of Multiple Linear Regression

2.6.8.1 Linearity There is a linear relationship between the dependent variable and each independent variable

Linearity may be evaluated by constructing a scatter diagram for each independent variable and examine the diagrams

Linearity can also be assessed graphically by constructing residual plots. Constructed by plotting residuals against the fitted values and this should exhibit no pattern

2.6.8.2 Homoscedasticity The variation in the residuals is the same for all fitted values of Y. The formal test for homoscedasticity is the Breusch Pagan test and the hypothesis is:

H_0 : Constant variance

H_a : Heteroscedasticity

2.6.8.3 Normality of residuals Residuals are normally distributed with a mean of zero. The assumption is necessary for the validity of the inferences that we make based on the global and individual hypothesis tests

The formal test for the normality of residuals is the Shapiro-Wilk test. The hypothesis tested is:

H_0 : Normality of residuals H_a : Residuals not normally distributed

2.6.8.4 Multicollinearity This exists when the independent variables are correlated. If an independent variable is highly correlated with other variables in the model should be removed.

To assess the degree to which independent variables are correlated we compute the VIF. A VIF greater than 10 is unsatisfactory.

2.6.8.5 Autocorrelation Successive residuals should be independent implying that there is no pattern in the residuals.

When successive residuals are correlated we refer to the condition as autocorrelation.

The formal test is the Durbin Watson test

H_0 : No Autocorrelation

H_a : Autocorrelation

2.6.9 Example in R

We fit a multiple linear regression for US consumption given by:

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \beta_3 X_3 + \beta_4 X_4$$

Where:

Y is the percentage change in real personal consumption expenditure X_1 is the percentage change in real personal disposable income X_2 is the percentage change in industrial production X_3 is the percentage change in personal savings X_4 is the change in unemployment rate

```
fit_model <- us_change |>
  model(tslm = TSLM(Consumption ~ Income + Production +
                    Unemployment + Savings))
report(fit_model)

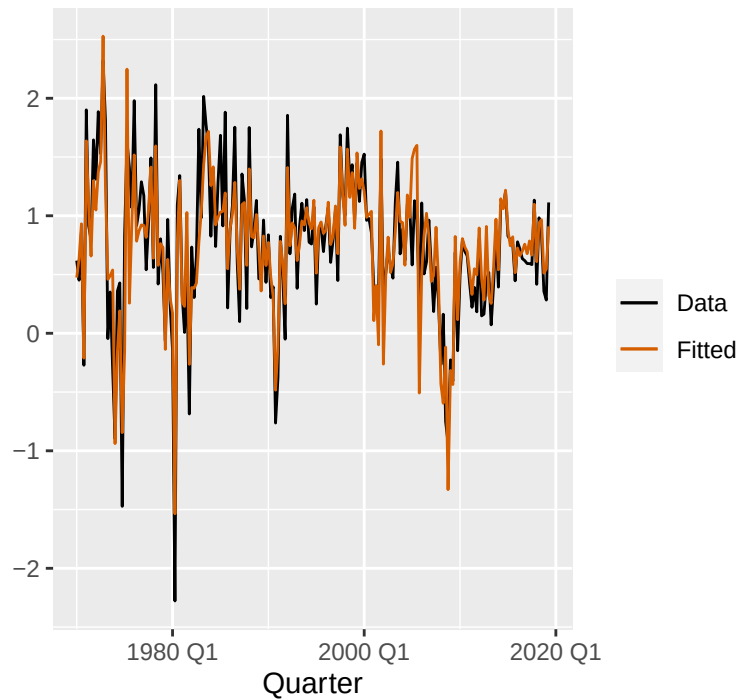
## Series: Consumption
## Model: TSLM
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.90555 -0.15821 -0.03608  0.13618  1.15471
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.253105   0.034470   7.343 5.71e-12 ***
## Income         0.740583   0.040115  18.461 < 2e-16 ***
## Production     0.047173   0.023142   2.038  0.0429 *
## Unemployment  -0.174685   0.095511  -1.829  0.0689 .
## Savings        -0.052890   0.002924 -18.088 < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.3102 on 193 degrees of freedom
## Multiple R-squared:  0.7683, Adjusted R-squared:  0.7635
## F-statistic: 160 on 4 and 193 DF, p-value: < 2.22e-16
```

The fitted values can be obtained as:

```
augment(fit_model) |>
  ggplot(aes(x = Quarter)) +
  geom_line(aes(y = Consumption, colour = "Data")) +
  geom_line(aes(y = .fitted, colour = "Fitted")) +
  labs(y = NULL,
       title = "Percent change in US consumption expenditure"
  ) +
```

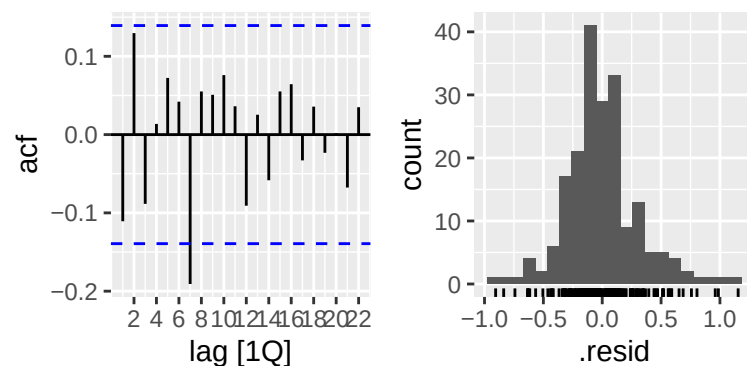
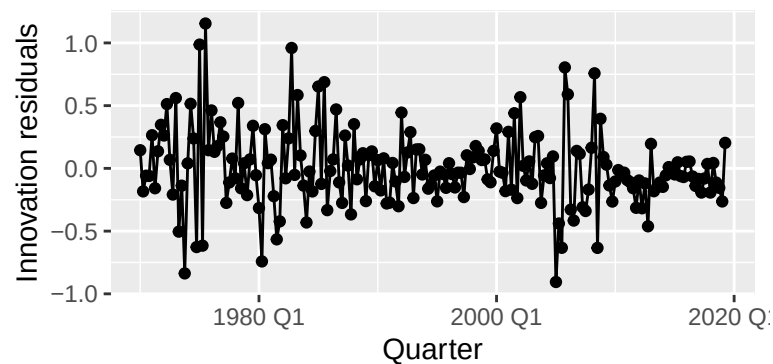
```
scale_colour_manual(values=c(Data="black",Fitted="#D55E00")) +  
guides(colour = guide_legend(title = NULL))
```

Percent change in US consumption expenditur



The histogram of residuals

```
fit_model |> gg_tsresiduals()
```



We fit the model and test the assumptions

```
model2 <- lm(Consumption ~ Income + Production +
              Unemployment + Savings, data = us_change)
pander(summary(model2))
```

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.2531	0.03447	7.343	5.713e-12
Income	0.7406	0.04012	18.46	1.648e-44
Production	0.04717	0.02314	2.038	0.04287
Unemployment	-0.1747	0.09551	-1.829	0.06895
Savings	-0.05289	0.002924	-18.09	2.028e-43

Table 10: Fitting linear model: Consumption ~ Income + Production + Unemployment + Savings

Observations	Residual Std. Error	R^2	Adjusted R^2
198	0.3102	0.7683	0.7635

The ANOVA table

```
pander(anova(model2))
```

Table 11: Analysis of Variance Table

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Income	1	11.8	11.8	122.6	2.215e-22
Production	1	15.67	15.67	162.9	1.921e-27
Unemployment	1	2.623	2.623	27.26	4.583e-07
Savings	1	31.48	31.48	327.2	2.028e-43
Residuals	193	18.57	0.09623	NA	NA

Testing the assumptions

Normality

```
check_normality(model2)
```

```
## Warning: Non-normality of residuals detected (p < .001).
```

Autocorrelation

```
check_autocorrelation(model2)
```

```
## OK: Residuals appear to be independent and not autocorrelated (p = 0.134).
```

Homoscedasticity

```
check_heteroscedasticity(model2)
```

```
## Warning: Heteroscedasticity (non-constant error variance) detected (p = 0.001).
```

Multicollinearity

```
check_collinearity(model2)
```

```
## # Check for Multicollinearity
```

```
##
```

```
## Low Correlation
```

```
##
```

```
##      Term  VIF   VIF 95% CI Increased SE Tolerance Tolerance 95% CI
##      Income 2.67 [2.18, 3.37]      1.63      0.37      [0.30, 0.46]
##      Production 2.54 [2.08, 3.19]      1.59      0.39      [0.31, 0.48]
##      Unemployment 2.52 [2.06, 3.17]      1.59      0.40      [0.32, 0.48]
##      Savings 2.51 [2.05, 3.15]      1.58      0.40      [0.32, 0.49]
```

Linearity

```
us_change |>
```

```
  ggpairs(columns = 2:6)
```

